

Midterm 2

Applied Algebra

April 15, 2025

Name: _____

ID #: _____

Instructions:

- No notes or books are allowed. Calculators are permitted.
- You must show your reasoning, not just the final answer. Answers without justification will receive only partial credit.
- If a problem asks you to solve an equation, you must find all solutions. If no solutions exist, you must explain why.
- Please cross out any work that is not part of your final solution (e.g., preliminary computations you didn't use).

Each problem is worth **25 points**.

	1	2	3	4	Total
Grade					

Exercise 1. (True/False) Justify your answers with a few sentences.

1. $7^{66} \equiv 49 \pmod{120}$.
2. The polynomial $x^3 - x^2 + x + 7$ has no integer root.
3. For any positive integer a , the last digit of a^5 is the same as that of a .
4. For every three sets A, B, C , we have $A \cup (B \cap C) = (A \cup B) \cap C$.
5. For any function $f: X \rightarrow Y$ and any subsets $A, B \subseteq X$, we have $f(A \setminus B) = f(A) \setminus f(B)$.
Here, $f(A) := \{f(a) : a \in A\}$.

Exercise 2. Chen has a bag of coins. After counting them, he discovered that when the coins are divided into 3 equal piles, there is 1 coin left over; when divided into 4 equal piles, there are 2 coins left over; and when divided into 5 equal piles, 3 coins remain.

- (a) Let n be the number of coins Chen had. Write the given information about n as a system of congruences.
- (b) What is the smallest possible number of coins Chen could have?

Exercise 3. The sequence (F_n) of Fibonacci numbers is defined by the conditions $F_1 = 1$, $F_2 = 1$, $F_{n+1} = F_n + F_{n-1}$ for all $n \geq 2$. Prove that for any $n \geq 1$, we have

$$F_1^2 + F_2^2 + \cdots + F_n^2 = F_n F_{n+1}.$$

Exercise 4. Find the greatest common divisor of $12n + 1$ and $30n + 2$, where n is a positive integer.